Data Modeling Automation: From Business Concept to Production-Ready Data Model

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Idea Title (Like a book, short and simple)

Data Modeling Automation: From Business Concept to Production-Ready Data Model

Select Your Theme

Automation and Operational Efficiency

What problem do you want to solve?

Data modeling is one of the most important steps in building trusted data products, but it remains highly manual, time-consuming, and dependent on a small number of experts.

Today, teams spend weeks translating business requirements into conceptual, logical, and physical data models. Business terms are interpreted differently across teams, documentation becomes outdated, and engineering teams often discover design issues late in delivery.

As a result:

• Data products take longer to deliver
• Modeling quality varies across teams
• Governance becomes difficult
• Reuse is limited
• AI initiatives are slowed by inconsistent data foundations

Why does it matter?

Every analytics platform, AI solution, operational workflow, and data product depends on a well-designed data model.

When data modeling is slow or inconsistent:

• Delivery timelines increase
• Technical debt accumulates
• Integration costs rise
• Data quality suffers

• AI adoption becomes more difficult
As organizations scale data products and AI capabilities, modeling cannot remain a manual craft performed one domain at a time.
It must become a repeatable, intelligent, and scalable capability.

Why is it challenging?

Data modeling sits at the intersection of business and technology.
The challenge is not simply generating tables and columns.
The challenge is understanding:
• Business concepts and definitions
• Relationships between entities
• Industry standards
• Data governance requirements
• Source system constraints
• Future extensibility
• Regulatory and compliance needs
Much of this knowledge exists in people's heads, PowerPoint decks, requirements documents, Jira stories, and legacy databases rather than in a reusable form.
Creating good models requires both technical expertise and domain expertise.

Why are you excited about it?

I'm excited because this represents one of the biggest opportunities to accelerate delivery across the enterprise.
Imagine describing a business capability:
"Create a longitudinal member record for risk and quality management."
And within minutes:
• A conceptual model is generated
• Logical entities and relationships are proposed
• Industry-standard attributes are suggested
• Existing enterprise assets are identified
• Data product contracts are created
• Metadata and lineage are automatically captured
• Engineering-ready specifications are produced

Instead of debating model design for weeks, teams begin with a high-quality first draft and focus their time on refinement and innovation.

This is not just productivity.

This is organizational scale.

What We'd Build During the Hackathon

An AI-Powered Data Model Designer.

The prototype would:

1. Natural Language to Data Model

Users describe a business capability:

"Build a quality reporting data product."

The platform automatically generates:

• Business entities
• Relationships
• Key attributes
• Conceptual model diagrams

2. Industry-Aware Modeling

Leverage healthcare standards and existing enterprise patterns to recommend:

• Member entities
• Provider entities
• Claims structures
• Clinical event structures
• Reference data

3. Intelligent Reuse Detection

Analyze existing data products and models to identify:

• Reusable entities
• Existing domains
• Duplicate modeling efforts
• Standard definitions

4. Automated Documentation

Generate:

• Data dictionaries
• Business definitions
• Data contracts
• Metadata descriptions
• Stewardship assignments

5. AI Design Assistant

Allow users to ask:

"What attributes are missing?"
"Will this support quality scoring?"
"How does this compare to our existing member model?"
"What downstream systems may be impacted?"

What do you need to make it happen? (who/skills/etc)

People

- Data architects
- Data modelers
- Product managers
- Domain SMEs
- Metadata and governance teams
- AI engineers

Data

- Existing enterprise data models
- Metadata repositories
- Data dictionaries
- Data contracts
- Business glossaries
- Healthcare standards and schemas

Technology

- LLM-based design assistant
- Metadata platform
- Data catalog
- Data modeling repository
- Knowledge graph
- Visualization tools

Expected Outcome

A platform that:

- Reduces modeling effort from weeks to hours
- Improves consistency across domains
- Accelerates data product delivery
- Increases reuse of enterprise assets
- Improves governance compliance
- Produces AI-ready data products by design

If you have a team identified, enter the Team Name

If you have a team, please list the name and email for each team member

I agree to flex my hours during the Hackathon Ideation Period

I Agree

I attest:

I attest that no members of this team are registered as a judge for this Hackathon, and all innovation, ideas, and effort put forth during this event must be original work. This team understands that all innovation, ideas and effort put forth during this event must be original

I Acknowledge and agree to the Hackathon Participant Agreement.